

EEE4114F Final Exam

June 2021

Duration: 3 hours

Total: 100 marks

Instructions:

- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has *six* questions totalling 60 marks. You must answer all of them.
- **Part B** has *three* questions totalling 40 marks. You must answer all of them.
- Part A and part B must be answered in separate exam books.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 3 hours.

Part A:

Question:	1	2	3	4	5	6	Total
Points:	10	10	10	10	10	10	60
Score:							

Part B:

Question:	7	8	9	Total
Points:	16	14	10	40
Score:				

Part A: Digital Signal Processing

Parts A and B must be answered in separate exam booklets.

Question 1 [10 marks]

Consider a LTI system with input $x[n]$, output $y[n]$, and an impulse response $h[n]$. For such a system the relationship $y[n] = h[n] * x[n]$ holds.

- a. Prove that as a consequence the relationship

$$\sum_{n=-\infty}^{\infty} y[n] = \left(\sum_{n=-\infty}^{\infty} h[n] \right) \left(\sum_{n=-\infty}^{\infty} x[n] \right)$$

holds for all LTI systems.

- b. Now assume that

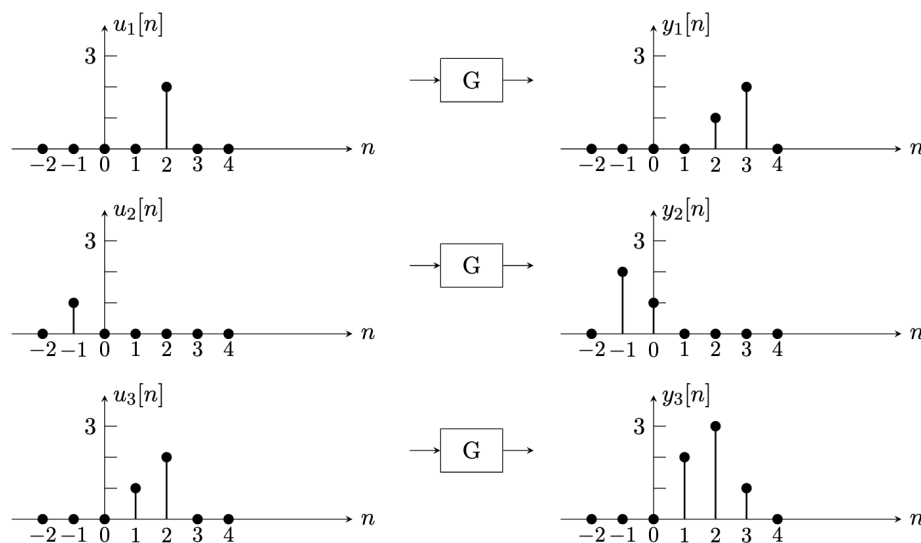
$$x[n] = \delta[n] + 2\delta[n-1] + \delta[n-2] \quad \text{and} \quad h[n] = 4\delta[n] - 2\delta[n-1] + \delta[n-2].$$

Compute and plot $y[n]$.

- c. Verify that your answer in (b) satisfies the test described in (a).

Question 2 [10 marks]

The system represented by G in the figure below is assumed to be time-invariant:

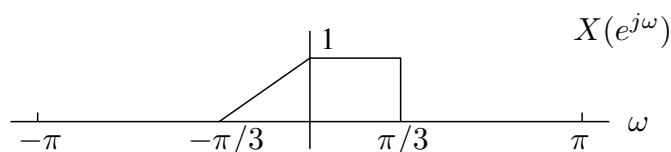


When the inputs to the system are $u_1[n]$, $u_2[n]$, and $u_3[n]$ the outputs are $y_1[n]$, $y_2[n]$, and $y_3[n]$ as shown.

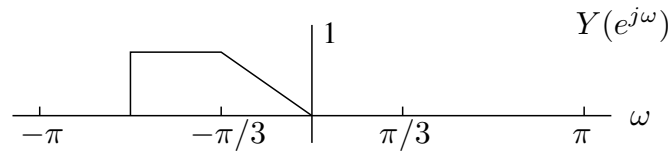
- a. Specify and plot the response of the system to the input $x[n] = \delta[n]$.
b. Show that the system is not linear.

Question 3 [10 marks]

A sequence $x[n]$ has a zero-phase DTFT $X(e^{j\omega})$ given below:



The sequence $y[n]$ has the following zero-phase DTFT:



Find an expression for $y[n]$ in terms of $x[n]$.

Question 4 [10 marks]

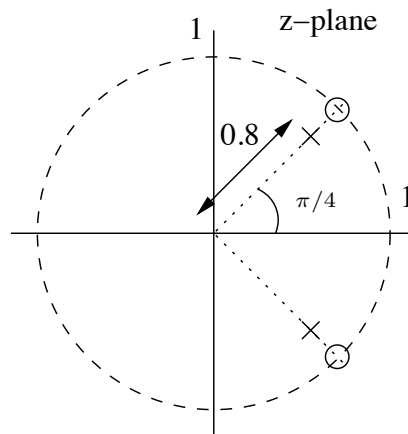
The continuous-time signal $x(t) = \sin(20\pi t) + \cos(40\pi t)$ is sampled with a sampling period T , with $t = nT$, to obtain the discrete-time signal

$$x[n] = \sin\left(\frac{\pi n}{5}\right) + \cos\left(\frac{2\pi n}{5}\right).$$

- Sketch the spectrum $X(e^{j\omega})$ of $x[n]$ over the range $-2\pi \leq \omega \leq 2\pi$.
- Determine a value of T consistent with this information.
- Is your choice of T in this last part unique? If so, explain why. If not, specify another choice of T consistent with the information given.

Question 5 [10 marks]

The pole-zero plot for a filter is as follows:



- Give an expression for the z-transform $H(z)$ of the system.
- Find the gain of the system for $\omega = 0$ and $\omega = \pi/4$ radians/sample.
- Use graphical methods to roughly sketch the magnitude response of the filter.
- Describe the effect the filter will have on an input signal.
- What are the possible regions of convergence for the filter? Which will correspond to a causal filter, and which to a stable filter?

Question 6 [10 marks]

A finite impulse response system has a linear phase response if for some N it satisfies either the even symmetric condition $h[n] = h[N-1-n]$ or the odd symmetric condition $h[n] = -h[N-1-n]$ for all n .

Consider the linear-phase finite impulse response filter described by the following LCCDE:

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2],$$

where $x[n]$ is the input and $y[n]$ the output, and b_0 , b_1 , and b_2 are real.

- a. Give the impulse response $h[n]$ of the filter in terms of the parameters b_0 , b_1 , and b_2 .
- b. Show that if $b_0 = b_2$ then the frequency response of the filter can be written in the form

$$H(e^{j\omega}) = A(e^{j\omega})e^{-j\omega}$$

for a purely real-valued function $A(e^{j\omega})$.

- c. Assume that the filter satisfies the even symmetry condition above for $N = 3$. Determine the values of the coefficients b_0 , b_1 , and b_2 such that the filter completely rejects frequency components at $\omega = 2\pi/3$ radians per sample and has unity gain at $\omega = 0$.

END OF PART A

Part B: Machine Learning

Parts A and B must be answered in separate exam booklets.

Question 7: Model evaluation [16 marks]

Your friend Geoff has been growing indoor plants using adjustable grow lights. He comes to you with a regression model that he claims is able to predict how much his plants will grow (in height) based the intensity of the light. He shows you the model's output in Figure 1 along with the target data he used to train the model with.

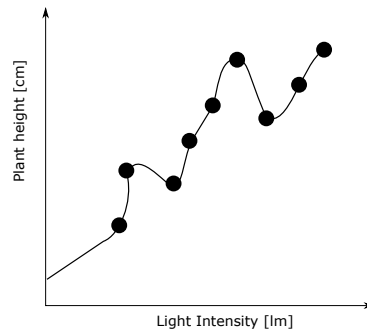


Figure 1: Geoff's model's output is shown by the solid line and the training data points indicated with black dots

- What did Geoff do wrong in terms of evaluating the performance of his model? (2)
- Is this model showing signs of over or under fitting? Explain your answer. (2)
- Explain why Geoff's model would perform poorly in a real world scenario. (3)
- Based on the model's output would you expect it to have a high or low number of trainable parameters? Explain your answer. (2)
- Geoff retrains his model again, but uses different sets of data he collected. This time his model learned very different parameters and the model outputs very different values than before. Is this an indication of high variance or high bias? Explain your answer. (2)
- Geoff collects more data, this time with far more input features. Many of these features are not particularly useful to the task. Which type of regularization penalty (l_1 or l_2) is likely to yield better results? (2)
- Explain what k-fold cross-validation is and why it would be a better method of evaluating how well this model generalizes to unseen data. (3)

Question 8: Neural Networks [14 marks]

You have been asked to develop a machine learning system to estimate the efficiency of a solar PV cell based on ambient temperature. It takes in a single input feature and outputs a single output value.

You come up with a fully-connected, feedforward neural network with two hidden layers. The first hidden layer has 2 neurons, and the second layer has one neuron. All of the activation functions for each neuron are Rectified Linear Units (ReLU).

The cost function you plan to optimize is given by: $J = \frac{1}{2}(\hat{\mathbf{y}} - \mathbf{y})^2$

Where $\hat{\mathbf{y}}$ is the predicted output of the model and $\mathbf{y}$ is the target variable.

- How many trainable parameters does this model have? (1)
- Suppose at a particular moment in time the input is given by $x = 3$, and the target value is given by $y = 10$, all of the biases are equal to 1, and the weight matrices have the following values.

$$W^1 = \begin{bmatrix} 3 & -2 \end{bmatrix} \quad W^2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- (i) Calculate the following values based on a single forward pass of the input.
- 1) $\mathbf{z}^1$, the net weighted inputs into all neurons in the first layer (1)
 - 2) $\mathbf{a}^1$, the activations of neurons in the first layer (1)
 - 3) $\mathbf{z}^2$, the net weighted inputs into all neurons in the second layer (1)
 - 4) $\mathbf{a}^2$, the activations of neurons in the second layer (1)
- (ii) After the forward pass has been performed your model parameters need to be updated. Calculate the following
- 1) δ^L , the errors in the last layer (3)
 - 2) δ^1 , the errors in the first hidden layer (2)
 - 3) $\frac{\partial J}{\partial \mathbf{W}^1}$, the gradients for weights in the first layer. (1)
 - 4) $\frac{\partial J}{\partial \mathbf{b}^1}$, the gradients for the biases in the first layer. (1)
- (iii) Using gradient descent calculate the new parameters for $\mathbf{W}^1$ and $\mathbf{b}^1$ given that $\eta = 0.1$ (2)

Question 9: Reinforcement Learning [10 marks]

You have a mobile robot in the discrete grid world shown in Figure 2. There are six states (locations on the grid) and four possible actions it can take at each time step by moving N/E/S/W provided there is space to move in to. s_4 and s_5 are terminal states with reward values of -1 and $+1$ respectively. The non-terminal states have zero reward values.

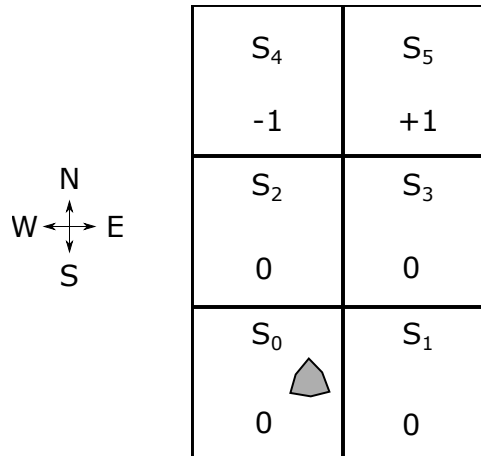


Figure 2: A mobile robot agent in a discrete grid world environment

You decide to use ϵ -greedy Q-learning to train a policy for the agent. All of the initial Q-values are set to zero. The Bellman equation below indicates how the Q-values are updated.

$$Q(s, a) \leftarrow Q(s, a) + \alpha(N(s, a))[R(s) + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Where $N(s, a)$ is the number of times the state-action pair (s, a) has been encountered. The learning rate, $\alpha(n)$ is inversely proportional to $N(s, a)$ such that $\alpha(n) = \frac{1}{n}$. The discount factor is set to $\gamma = 0.8$.

Figure 3 shows the $N(s, a)$ (indicates number of times (s, a) is encountered) and $Q(s, a)$ 'tables' for the second and third trials. The relative positions of the values in each state cell indicate the action chosen e.g. The number at the top of a cell indicates the action is 'N' and to the left of the cell is 'W' etc. This is to make visualising the transitions easier.

- a. Explain why $\alpha(n) = \frac{1}{n}$ could be beneficial compared to a fixed learning rate. (2)
- b. Write down the series of (state, reward, actions) taken at t_2 that produced the $N(s, a)$ and $Q(s, a)$ tables for t_3 as shown in Figure 3. (1)
- c. At t_3 the following transitions are performed by the action selection policy: (3)

$$(s_0, 0, E) \longrightarrow (s_1, 0, N) \longrightarrow (s_3, 0, N) \longrightarrow (s_5, 1)$$

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with } W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$